

Lab protocol Day 3 – editing experiment

Today we perform the actual editing experiment. We prepare electrocompetent *E. coli* cells and pre-induce the Red recombineering system, transform the gRNA and linear donor to start editing, and then plate cells on selective media for the gRNA plasmid and editing plasmid.

Induce expression of Lambda Red genes and prepare electrocompetent cells (4 hours)

Materials: 20% arabinose solution (sterile), pre-chilled sterile water, cuvettes, 1x 100 ml flask, LB + ampicillin, 50 ml falcon tube (pre-chilled), serological pipettes (25 ml)

Equipment: Bunsen burner, OD meter, bucket with ice, centrifuge, shaking incubator

1. Dilute overnight culture 1:10 in cuvette and measure OD₆₀₀. If OD₆₀₀ of 1:10 dilution is 0.45-0.50 go to step 2. Otherwise, dilute overnight culture until you reach OD₆₀₀ of 0.45-0.50 of the 1:10 dilution.
2. Dilute overnight culture 1:100 in LB + ampicillin. Grow at 37°C, shaking to OD₆₀₀ 0.4-0.6 (ca. 2 hours). *For 4-6 transformations, dilute 250 µL overnight in 25 ml LB + ampicillin*
3. Once culture has reached OD 0.4, add arabinose to final concentration of 0.75% to induce Lambda Red genes. Return flask to 37°C shaking incubator for 30 minutes. *For 25 ml culture: 938 µL 20% arabinose*
4. Immediately after 30 minutes induction, place the flask on ice to rapidly cool. Chill on ice 30 minutes.
5. Transfer 15 mL culture to a pre-chilled 50 mL falcon tube on ice.
6. Centrifuge at 4000 rpm for 5 minutes.
7. Pour off supernatant and place tube on ice.
8. Add 30 mL pre-chilled ice-cold sterile water to the tube. Gently resuspend the cells.
9. Spin as before and pour off the supernatant.
10. Add 1 mL ice-cold sterile water to the tube. Gently resuspend the cells.
11. Spin as before and pour off the supernatant.
12. Resuspend pellet in 200 µL ice-cold sterile water. Continue with transformation immediately!

Miniprep correct gRNA plasmid (30 min) (during incubation step between 2 and 3 above):

Materials: 2 Eppendorf tubes, Qiagen kit, Sterile water, Qubit tubes + dsDNA BR solution

Equipment: tabletop centrifuge, Qubit device

1. Take culture for which sequencing was correct and perform miniprep following protocol provided by supplier (*QIAprep_Miniprep.pdf*, protocol is on p. 26 onwards).
2. Measure concentration with Qubit (same protocol as Day 1) and dilute gRNA plasmid to 25 ng/µL.

Transform gRNA plasmid and linear donor (2 hours):

Materials: gRNA plasmid, donor template, labelled Eppendorf tube on ice, 2 pre-chilled labelled electroporation cuvettes (1mm gap width), labelled Eppendorf tube at RT for recovery with SOC + ampicillin + 0.75% arabinose, McConkey agar plates + amp + spec

Equipment: Bunsen burner, electroporator, heat block

1. Transfer 2 µL of the 25 ng/µL gRNA plasmid to two Eppendorf tubes on ice.
 - a. Add 50 ng of the PTC donor to one tube
 - b. Add 90.1 ng of KO donor to other tube
2. Add 50 µL competent cells to the Eppendorf tubes and tap gently to mix.
3. Transfer contents of Eppendorf tubes to pre-chilled electroporation cuvette.

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4. Prepare 1 mL SOC + ampicillin + 0.75% arabinose per transformation to select for editing plasmid in Eppendorf tube.
5. Turn on electroporation device and set to Ec1. Turn on flame.
6. Working by the flame, remove cap of electroporation cuvette, wipe off ice, and place in the holder. Press the “pulse” button and check that the time constant is around 5.50 ms. Write down time constants.
7. Immediately add 200 µL SOC + ampicillin + 0.75% arabinose from Eppendorf tube to the cuvette, mix gently by pipetting and transfer contents to the remaining 800 µL in the Eppendorf tube.
8. Recover for 1 hour at 37°C, shaking.
9. Make x-fold dilutions (see table) per setup in sterile Eppendorf tubes
 - a. 1/50: dilute 4 µL sample in 196 µL LB
 - b. 1/100: dilute 2 µL sample in 198 µL LB
 - c. 1/1000: dilute 20 µL of 1/100 dilution in 180 µL LB
 - d. 1/5000: dilute 40 µL of 1/1000 dilution in 160 µL LB
 - e. 1/10000: dilute 20 µL of 1/1000 dilution in 180 µL LB
10. Plate 100 µL of each dilution on McConkey agar plates + amp + spec to select for Cas9 and guide/donor plasmids.
11. Grow overnight at 37°C.

Sample	gRNA	donor	Dilutions to plate	Groups
1.1	gRNA1	PTC donor gRNA 1	50	1 & 4
1.2	gRNA1	PTC donor gRNA 1	100	1 & 4
1.3	gRNA1	PTC donor gRNA 1	1000	1 & 4
2.1	gRNA2	PTC donor gRNA 2	1000	2 & 5
2.2	gRNA2	PTC donor gRNA 2	5000	2 & 5
2.3	gRNA2	PTC donor gRNA 2	10000	2 & 5
3.1	gRNA3	PTC donor gRNA 3	500	3 & 6
3.2	gRNA3	PTC donor gRNA 3	1000	3 & 6
3.3	gRNA3	PTC donor gRNA 3	10000	3 & 6
4.1	gRNA1	KO donor	1000	1 & 4
4.2	gRNA1	KO donor	5000	1 & 4
4.3	gRNA1	KO donor	10000	1 & 4
5.1	gRNA2	KO donor	1000	2 & 5
5.2	gRNA2	KO donor	5000	2 & 5
5.3	gRNA2	KO donor	10000	2 & 5
6.1	gRNA3	KO donor	1000	3 & 6
6.2	gRNA3	KO donor	5000	3 & 6
6.3	gRNA3	KO donor	10000	3 & 6